Hyperelastic Material, Potential & Complementary Strain Energy

Introduction

We have utilized the constitutive law $\boldsymbol{\sigma} = \boldsymbol{C}:\boldsymbol{\epsilon}$ without proper explanation in the previous chapter. In this chapter, we shall introduce the hyperelastic material behind the constitutive law and its relationship to work and energy.

While energy has many other physical interpretations, the energy of a path independent object/system contains the complete information to describe the current stage of the object/system.

Hyperelastic Material & Potential Strain Energy Density

General Hyperelastic Material

The hyperelastic material is the material that has path independence of the stress, means if strains $\boldsymbol{\epsilon}_1 = \boldsymbol{\epsilon}_2$, then $\boldsymbol{\sigma}(\boldsymbol{\epsilon}_1) = \boldsymbol{\sigma}(\boldsymbol{\epsilon}_2)$ no matter how to achieve this strain.

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = \frac{\partial v_\epsilon(\boldsymbol{\epsilon})}{\partial \boldsymbol{\epsilon}}$$

Where the v_ϵ is the potential strain energy density:

$$(informal, replace\ variation\ by\ differential\ \boldsymbol{\delta\epsilon} = \boldsymbol{d\epsilon}): v_\epsilon(\boldsymbol{\epsilon}+\boldsymbol{h}) - v_\epsilon(\boldsymbol{h}) = \int_{\boldsymbol{h}}^{\boldsymbol{\epsilon}+\boldsymbol{h}} \boldsymbol{\sigma}(\boldsymbol{\epsilon}):\boldsymbol{d\epsilon}$$

$$(formal, only\ valid\ if\ path\ independence): v_\epsilon(\boldsymbol{\epsilon}+\boldsymbol{h}) - v_\epsilon(\boldsymbol{\epsilon}) = \int_0^1 \boldsymbol{\sigma}(t\boldsymbol{\epsilon}+\boldsymbol{h}):\boldsymbol{\epsilon}dt$$

The elasticity (stiffness) tensor $\boldsymbol{C}$ can then be defined as:

$$\boldsymbol{C} = \frac{\partial \boldsymbol{\sigma}(\boldsymbol{\epsilon})}{\partial \boldsymbol{\epsilon}} = \frac{\partial^2 v_\epsilon(\boldsymbol{\epsilon})}{\partial \boldsymbol{\epsilon}^2}$$

$\boldsymbol{C}$ is a fourth-order tensor (check tensor derivative chapter):

$$\boldsymbol{C} = C_{ijkl}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l$$

It has a total of 3x3x3x3=81 components $\boldsymbol{C} \sim C_{ijkl}$. However, not all of these are independent. Since stress and strain tensors are symmetric, there are only six independent components of each. Thus, instead of 9x9=81 components, the elasticity tensor has only 6x6=36 independent components.

$$(\epsilon_{kl} = \epsilon_{lk}) \quad \boldsymbol{\sigma} = \boldsymbol{C}:\boldsymbol{\epsilon} \sim \sigma_{ij} = C_{ijkl}\epsilon_{kl} = C_{ijlk}\epsilon_{kl} \Rightarrow C_{ijkl} = C_{ijlk}$$

$$(\sigma_{ij} = \sigma_{ji}) \quad \boldsymbol{\sigma} = \boldsymbol{C}:\boldsymbol{\epsilon} \sim \sigma_{ij} = C_{ijkl}\epsilon_{kl} = C_{jikl}\epsilon_{kl} \Rightarrow C_{ijkl} = C_{jikl}$$

Furthermore, since the order of differentiation of the strain energy function with respect to the components of the strain tensor is immaterial, the elasticity tensor is symmetric with respect o ij and kl. A symmetric six by six matrix has only 21 independent terms.

$$v_\epsilon(\boldsymbol{\epsilon}) = \int_0^1 \boldsymbol{\sigma}(t\boldsymbol{\epsilon}) : \boldsymbol{\epsilon}\, dt = \int_0^1 \boldsymbol{\epsilon} : \boldsymbol{C} : \boldsymbol{\epsilon}\, dt = \frac{1}{2}\boldsymbol{\epsilon} : \boldsymbol{C} : \boldsymbol{\epsilon}$$

$$\boldsymbol{C} = \frac{\partial^2 v_\epsilon(\boldsymbol{\epsilon})}{\partial \boldsymbol{\epsilon}^2} \sim C_{abcd} = \frac{1}{2}\frac{\partial^2\left(\epsilon_{ij} C_{ijkl} \epsilon_{kl}\right)}{\partial \epsilon_{ab} \partial \epsilon_{cd}} = \frac{1}{2}(C_{abcd} + C_{cdab}) \Rightarrow C_{abcd} = C_{cdab}$$

Isotropy

Material is said to be isotropic if its properties do not depend upon certain preferential directions, it can also be said that the elasticity tensor be invariant with respect to coordinate transformation in the language of tensor:

$$v_\epsilon(\boldsymbol{\epsilon}) = v_\epsilon\left(\boldsymbol{f_1}(\boldsymbol{\epsilon}) + \boldsymbol{f_2}(\boldsymbol{\epsilon}) + \boldsymbol{f_3}(\boldsymbol{\epsilon})\right)$$

Where:

$$f_1(\boldsymbol{\epsilon}) = tr(\boldsymbol{\epsilon}) = \epsilon_{ii}$$

$$f_2(\boldsymbol{\epsilon}) = tr(\boldsymbol{\epsilon}^2) = \epsilon_{ij}\epsilon_{ji}$$

$$f_3(\boldsymbol{\epsilon}) = tr(\boldsymbol{\epsilon}^3) = \epsilon_{ij}\epsilon_{jk}\epsilon_{ki}$$

The stress tensor $\boldsymbol{\sigma}$ is then:

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = \frac{\partial v_\epsilon(\boldsymbol{\epsilon})}{\partial \boldsymbol{\epsilon}} = \frac{\partial v_\epsilon}{\partial f_1}\frac{\partial f_1}{\partial \boldsymbol{\epsilon}} + \frac{\partial v_\epsilon}{\partial f_2}\frac{\partial f_2}{\partial \boldsymbol{\epsilon}} + \frac{\partial v_\epsilon}{\partial f_3}\frac{\partial f_3}{\partial \boldsymbol{\epsilon}}$$

$$\frac{\partial f_1}{\partial \boldsymbol{\epsilon}} = \frac{\partial \epsilon_{kk}}{\partial \epsilon_{ij}} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \sim \delta_{ki}\delta_{kj} = \delta_{ij} \Rightarrow \boldsymbol{I}$$

$$\frac{\partial f_2}{\partial \boldsymbol{\epsilon}} = \frac{\partial \epsilon_{kl}\epsilon_{lk}}{\partial \epsilon_{ij}} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \sim \delta_{ki}\delta_{lj}\epsilon_{lk} + \epsilon_{kl}\delta_{li}\delta_{kj} = 2\epsilon_{ji} = 2\epsilon_{ij} \Rightarrow \boldsymbol{2\epsilon}$$

$$\frac{\partial f_3}{\partial \boldsymbol{\epsilon}} = \frac{\partial \epsilon_{kl}\epsilon_{lm}\epsilon_{mk}}{\partial \epsilon_{ij}} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \sim \delta_{ki}\delta_{lj}\epsilon_{lm}\epsilon_{mk} + \epsilon_{kl}\delta_{li}\delta_{mj}\epsilon_{mk} + \epsilon_{kl}\epsilon_{lm}\delta_{mi}\delta_{kj} = \epsilon_{jm}\epsilon_{mi} + \epsilon_{ki}\epsilon_{jk} + \epsilon_{jl}\epsilon_{li}$$
$$= 3\epsilon_{jk}\epsilon_{ki} = 3\epsilon_{ik}\epsilon_{kj} \Rightarrow \boldsymbol{3\epsilon^2}$$

Then:

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = \frac{\partial v_\epsilon(\boldsymbol{\epsilon})}{\partial \boldsymbol{\epsilon}} = \frac{\partial v_\epsilon}{\partial f_1}\boldsymbol{I} + \frac{\partial v_\epsilon}{\partial f_2}\boldsymbol{2\epsilon} + \frac{\partial v_\epsilon}{\partial f_3}\boldsymbol{3\epsilon^2}$$

Note:

$$\boldsymbol{E} = \boldsymbol{E}^T$$

$$(\boldsymbol{A} \cdot \boldsymbol{B})^T = \boldsymbol{B}^T \cdot \boldsymbol{A}^T$$

$$(\boldsymbol{E} \cdot \boldsymbol{E})^T = \boldsymbol{E}^T \cdot \boldsymbol{E}^T = \boldsymbol{E} \cdot \boldsymbol{E}$$

Linear, Isotropic Elasticity

If we wish to have a linear isotropic constitutive law, the potential strain energy v_ϵ must be a purely quadratic function of strain, and it can depend only on f_1^2 and f_2 (f_1 is only linear in the components of $\boldsymbol{\epsilon}$, while f_3 is cubic in the components of $\boldsymbol{\epsilon}$):

$$v_\epsilon(\boldsymbol{\epsilon}) = a_1 f_1^2(\boldsymbol{\epsilon}) + a_2 f_2(\boldsymbol{\epsilon})$$

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = \frac{\partial v_\epsilon(\boldsymbol{\epsilon})}{\boldsymbol{\partial \epsilon}} = 2a_1 f_1 \boldsymbol{I} + a_2 \boldsymbol{\epsilon}$$

Written in our familiar form:

$$\boldsymbol{\sigma} = \lambda tr(\boldsymbol{\epsilon})\boldsymbol{I} + 2\mu\boldsymbol{\epsilon}$$

$$\boldsymbol{\epsilon} = -\frac{\lambda}{2\mu(3\lambda + 2\mu)} tr(\boldsymbol{\sigma})\boldsymbol{I} + \frac{1}{2\mu}\boldsymbol{\sigma}$$

The two constants λ, μ are called Lamé parameters. In this review we shall use another set of parameters E, v

$$\boldsymbol{\sigma} = \frac{E}{1+v}[\frac{v}{1-2\mu} tr(\boldsymbol{\epsilon})\boldsymbol{I} + \boldsymbol{\epsilon}]$$

$$\boldsymbol{\epsilon} = -\frac{v}{E} tr(\boldsymbol{\sigma})\boldsymbol{I} + \frac{1+v}{E}\boldsymbol{\sigma}$$

Potential Strain Energy

We have briefly introduced the potential strain energy density function while defining the hyperelastic material, as described in the introduction, the potential strain energy density contains the complete information to describe the current stage of the point where the density is based upon:

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = \frac{\partial v_\epsilon(\boldsymbol{\epsilon})}{\boldsymbol{\partial \epsilon}}$$

The potential strain energy of a solid body is defined as the integral of potential strain energy density function over the entire material domain; it is a functional:

$$V_\epsilon(\boldsymbol{\epsilon}) = \int v_\epsilon(\boldsymbol{\epsilon})\, d\Omega$$

If the derivative of v_ϵ would produce $\boldsymbol{\sigma}$ directly, what would be given by the "derivative" of V_ϵ. Combining what we have discussed about the functional in variational principle chapter and the directional derivative in tensor derivatives chapter, we have:

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = DV_\epsilon(\boldsymbol{\epsilon}; \boldsymbol{\delta\epsilon}) = \int Dv_\epsilon(\boldsymbol{\epsilon}; \boldsymbol{\delta\epsilon})\, d\Omega = \int \frac{\partial v_\epsilon}{\partial \boldsymbol{\epsilon}} : \boldsymbol{\delta\epsilon}\, d\Omega = \int \boldsymbol{\sigma}(\boldsymbol{\epsilon}) : \boldsymbol{\delta\epsilon}\, d\Omega$$

Utilizing:

$$\boldsymbol{\epsilon}(\boldsymbol{u}) = \frac{1}{2}(\boldsymbol{u} \otimes \boldsymbol{\nabla} + \boldsymbol{\nabla} \otimes \boldsymbol{u})$$

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$$

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = \int \boldsymbol{\sigma}(\boldsymbol{\epsilon}) : grad(\boldsymbol{\delta u})\, d\Omega = \int grad(\boldsymbol{\delta u}) : \boldsymbol{\sigma}(\boldsymbol{\epsilon})\, d\Omega$$

Note that:

$$grad(\boldsymbol{\delta u}) : \boldsymbol{\sigma}(\boldsymbol{\epsilon}) = div\big(\boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon})\big) - \boldsymbol{\delta u} \cdot div\big(\boldsymbol{\sigma}(\boldsymbol{\epsilon})\big)$$

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = \int div\big(\boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon})\big)\, d\Omega - \int \boldsymbol{\delta u} \cdot div\big(\boldsymbol{\sigma}(\boldsymbol{\epsilon})\big)\, d\Omega$$

Verify:

$$div(\boldsymbol{\sigma}) = \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} = \frac{\partial \sigma_{ij}}{\partial x_j} \boldsymbol{e}_i$$

$$\frac{\partial \delta u_i}{\partial x_j} \sigma_{ij} = \frac{\partial \delta u_i \sigma_{ij}}{\partial x_j} - \delta u_i \frac{\partial \sigma_{ij}}{\partial x_j}$$

Utilizing:

$$div(\boldsymbol{\sigma}) + \boldsymbol{f} = \boldsymbol{0}$$

$$-\int \boldsymbol{\delta u} \cdot div\big(\boldsymbol{\sigma}(\boldsymbol{\epsilon})\big)\, d\Omega = \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega$$

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = \int div\big(\boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon})\big)\, d\Omega + \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega$$

Divergence theorem (right divergence):

$$\int div(\boldsymbol{S})\, d\Omega = \int \boldsymbol{S} \cdot \boldsymbol{\nabla}\, d\Omega = \int \boldsymbol{S} \cdot \boldsymbol{n}\, d\partial\Omega$$

$$\int div\big(\boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon})\big)\, d\Omega = \int \boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon}) \cdot \boldsymbol{n}\, d\partial\Omega$$

Finally:

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = \int \boldsymbol{\sigma}(\boldsymbol{\epsilon}) : \boldsymbol{\delta\epsilon}\, d\Omega = \int \boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon}) \cdot \boldsymbol{n}\, d\partial\Omega + \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega$$

If boundary conditions are further considered:

$$\begin{cases} \boldsymbol{u} = \boldsymbol{h} \quad on\ \Gamma_h \\ \boldsymbol{\sigma} \cdot \boldsymbol{n} = \boldsymbol{q} \quad on\ \Gamma_q \\ \Gamma_h + \Gamma_q = \partial\Omega \end{cases}$$

Further express the stress on the displacement boundary (essential boundary):

$$\boldsymbol{\sigma} \cdot \boldsymbol{n} = \boldsymbol{p} \quad on\ \Gamma_h$$

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = \int \boldsymbol{\sigma}(\boldsymbol{\epsilon}) : \boldsymbol{\delta\epsilon}\, d\Omega = \int \boldsymbol{\delta u} \cdot \boldsymbol{p}\, d\Gamma_h + \int \boldsymbol{\delta u} \cdot \boldsymbol{q}\, d\Gamma_q + \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega$$

The variation of potential strain energy reveals the relationship between the internal response of an elastic body and the external action in terms of force/stress.

Complementary Strain Energy

Since the potential strain energy is introduced by integrating two field variables $\boldsymbol{\sigma}, \boldsymbol{\epsilon}$ as function of $\boldsymbol{\epsilon}$ over stages, it is also natural to do this in an interchangeable way:

$$\boldsymbol{\epsilon}(\boldsymbol{\sigma}) = \frac{\partial v_c(\boldsymbol{\sigma})}{\boldsymbol{\partial\sigma}}$$

Where v_c is the complementary strain energy density:

$$(informal, replace\ variation\ by\ differential\ \boldsymbol{\delta\sigma} = \boldsymbol{d\sigma}) : v_c(\boldsymbol{\sigma} + \boldsymbol{q}) - v_c(\boldsymbol{q}) = \int_{\boldsymbol{q}}^{\boldsymbol{\sigma}+\boldsymbol{q}} \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{d\sigma}$$

$$(formal, only\ valid\ if\ path\ independence) : v_c(\boldsymbol{\sigma} + \boldsymbol{q}) - v_c(\boldsymbol{q}) = \int_0^1 \boldsymbol{\epsilon}(t\boldsymbol{\sigma} + \boldsymbol{q}) : \boldsymbol{\sigma} dt$$

The flexibility tensor $\boldsymbol{D}$ can then be defined as:

$$\boldsymbol{D} = \frac{\partial \boldsymbol{\epsilon}(\boldsymbol{\sigma})}{\partial \boldsymbol{\sigma}} = \frac{\partial^2 v_c(\boldsymbol{\sigma})}{\partial \boldsymbol{\sigma}^2}$$

$\boldsymbol{D}$ is also a fourth-order tensor (check tensor derivative chapter):

$$\boldsymbol{D} = D_{ijkl}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l$$

And share the same symmetry as $\boldsymbol{C}$:

$$D_{ijkl} = D_{ijlk} = D_{jikl} = D_{klij}$$

The complementary strain energy of a solid body is defined as the integral of potential strain energy density function over the entire material domain; it is a functional:

$$V_c(\boldsymbol{\sigma}) = \int v_c(\boldsymbol{\sigma})\, d\Omega$$

Similarly to V_ϵ, if the derivative of v_c would produce $\boldsymbol{\epsilon}$ directly, what would be given by the "derivative" of V_c

$$\delta V_c(\boldsymbol{\sigma}) = DV_c(\boldsymbol{\sigma}; \boldsymbol{\delta\sigma}) = \int Dv_c(\boldsymbol{\sigma}; \boldsymbol{\delta\sigma})\, d\Omega = \int \frac{\partial v_c}{\partial \boldsymbol{\sigma}} : \boldsymbol{\delta\sigma}\, d\Omega = \int \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{\delta\sigma}\, d\Omega$$

Following the similar procedure (except the use of equilibrium $div(\boldsymbol{\sigma}) + \boldsymbol{f} = \boldsymbol{0}$):

$$\delta V_c(\boldsymbol{\sigma}) = \int \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{\delta\sigma}\, d\Omega = \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\partial\Omega - \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot div(\boldsymbol{\delta\sigma})\, d\Omega$$

If boundary conditions are further considered:

$$\begin{cases} \boldsymbol{u} = \boldsymbol{h} \quad on\ \Gamma_h \\ \boldsymbol{\sigma} \cdot \boldsymbol{n} = \boldsymbol{q} \quad on\ \Gamma_q \\ \Gamma_h + \Gamma_q = \partial\Omega \end{cases}$$

Further express the displacement on the stress boundary (natural boundary):

$$\boldsymbol{u} = \boldsymbol{g} \quad on\ \Gamma_q$$

$$\delta V_c(\boldsymbol{\sigma}) = \int \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{\delta\sigma}\, d\Omega = \int \boldsymbol{h} \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{g} \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_q - \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot div(\boldsymbol{\delta\sigma})\, d\Omega$$

The variation of complementary strain energy reveals the relationship between the internal response of an elastic body and the external action in terms of displacement.